

300 Physics Multiple Choice Questions

100 Simple, 100 Normal, and 100 Hard Questions with Answers

Comprehensive Physics Question Bank

Part 1: Simple Questions (1 - 100)

1. What is the SI unit of force? (Q1)

- A) Joule B) Newton C) Watt D) Pascal

Answer: B

2. Which is a scalar quantity? (Q2)

- A) Velocity B) Force C) Speed D) Acceleration

Answer: C

3. Value of acceleration due to gravity (g)? (Q3)

- A) 8.8 m/s² B) 9.8 m/s² C) 10.5 m/s² D) 11.2 m/s²

Answer: B

4. Instrument used to measure electric current? (Q4)

- A) Voltmeter B) Ammeter C) Ohmmeter D) Galvanometer

Answer: B

5. Energy possessed by a body due to motion? (Q5)

- A) Potential B) Kinetic C) Thermal D) Chemical

Answer: B

6. Newton's third law states action and reaction are: (Q6)

- A) Equal & same B) Unequal & opposite C) Equal & opposite D) Zero

Answer: C

7. SI unit of electrical resistance? (Q7)

- A) Volt B) Ampere C) Ohm D) Watt

Answer: C

8. Device converting electrical energy to mechanical? (Q8)

- A) Generator B) Motor C) Transformer D) Resistor

Answer: B

9. Wave type that requires a medium? (Q9)

- A) Light B) Radio C) Sound D) X-ray

Answer: C

10. Boiling point of water in Celsius? (Q10)

- A) 0°C B) 50°C C) 100°C D) 212°C

Answer: C

11. SI unit of power? (Q11)

- A) Joule B) Watt C) Volt D) Ampere

Answer: B

12. Type of lens thinner at edges, thicker in middle? (Q12)

- A) Concave B) Convex C) Plano D) Cylindrical

Answer: B

13. Freezing point of water in Fahrenheit? (Q13)

- A) 0°F B) 32°F C) 100°F D) 212°F

Answer: B

14. Color of light that bends most through a prism? (Q14)

- A) Red B) Yellow C) Green D) Violet

Answer: D

15. SI unit of frequency? (Q15)

- A) Second B) Hertz C) Meter D) Decibel

Answer: B

16. Which pole of a magnet attracts the north pole of another? (Q16)

- A) North pole B) South pole C) Both D) Neither

Answer: B

17. What is the center of a lens called? (Q17)

- A) Focus B) Optical center C) Pole D) Radius

Answer: B

18. Sound travels fastest in which medium? (Q18)

- A) Air B) Water C) Steel D) Vacuum

Answer: C

19. What does 'g' stand for in physics equations? (Q19)

- A) Gram B) Gravity acceleration C) Gauss D) Giga

Answer: B

20. Which mirror produces virtual and diminished images only? (Q20)

- A) Plane B) Concave C) Convex D) Parabolic

Answer: C

21. What is the SI unit of force? (Q21)

- A) Joule B) Newton C) Watt D) Pascal

Answer: B

22. Which is a scalar quantity? (Q22)

- A) Velocity B) Force C) Speed D) Acceleration

Answer: C

23. Value of acceleration due to gravity (g)? (Q23)

- A) 8.8 m/s² B) 9.8 m/s² C) 10.5 m/s² D) 11.2 m/s²

Answer: B

24. Instrument used to measure electric current? (Q24)

- A) Voltmeter B) Ammeter C) Ohmmeter D) Galvanometer

Answer: B

25. Energy possessed by a body due to motion? (Q25)

- A) Potential B) Kinetic C) Thermal D) Chemical

Answer: B

26. Newton's third law states action and reaction are: (Q26)

- A) Equal & same B) Unequal & opposite C) Equal & opposite D) Zero

Answer: C

27. SI unit of electrical resistance? (Q27)

- A) Volt B) Ampere C) Ohm D) Watt

Answer: C

28. Device converting electrical energy to mechanical? (Q28)

- A) Generator B) Motor C) Transformer D) Resistor

Answer: B

29. Wave type that requires a medium? (Q29)

- A) Light B) Radio C) Sound D) X-ray

Answer: C

30. Boiling point of water in Celsius? (Q30)

- A) 0°C B) 50°C C) 100°C D) 212°C

Answer: C

31. SI unit of power? (Q31)

- A) Joule B) Watt C) Volt D) Ampere

Answer: B

32. Type of lens thinner at edges, thicker in middle? (Q32)

- A) Concave B) Convex C) Plano D) Cylindrical

Answer: B

33. Freezing point of water in Fahrenheit? (Q33)

- A) 0°F B) 32°F C) 100°F D) 212°F

Answer: B

34. Color of light that bends most through a prism? (Q34)

- A) Red B) Yellow C) Green D) Violet

Answer: D

35. SI unit of frequency? (Q35)

- A) Second B) Hertz C) Meter D) Decibel

Answer: B

36. Which pole of a magnet attracts the north pole of another? (Q36)

- A) North pole B) South pole C) Both D) Neither

Answer: B

37. What is the center of a lens called? (Q37)

- A) Focus B) Optical center C) Pole D) Radius

Answer: B

38. Sound travels fastest in which medium? (Q38)

- A) Air B) Water C) Steel D) Vacuum

Answer: C

39. What does 'g' stand for in physics equations? (Q39)

- A) Gram B) Gravity acceleration C) Gauss D) Giga

Answer: B

40. Which mirror produces virtual and diminished images only? (Q40)

- A) Plane B) Concave C) Convex D) Parabolic

Answer: C

41. What is the SI unit of force? (Q41)

- A) Joule B) Newton C) Watt D) Pascal

Answer: B

42. Which is a scalar quantity? (Q42)

- A) Velocity B) Force C) Speed D) Acceleration

Answer: C

43. Value of acceleration due to gravity (g)? (Q43)

- A) 8.8 m/s² B) 9.8 m/s² C) 10.5 m/s² D) 11.2 m/s²

Answer: B

44. Instrument used to measure electric current? (Q44)

- A) Voltmeter B) Ammeter C) Ohmmeter D) Galvanometer

Answer: B

45. Energy possessed by a body due to motion? (Q45)

- A) Potential B) Kinetic C) Thermal D) Chemical

Answer: B

46. Newton's third law states action and reaction are: (Q46)

- A) Equal & same B) Unequal & opposite C) Equal & opposite D) Zero

Answer: C

47. SI unit of electrical resistance? (Q47)

- A) Volt B) Ampere C) Ohm D) Watt

Answer: C

48. Device converting electrical energy to mechanical? (Q48)

A) Generator B) Motor C) Transformer D) Resistor

Answer: B

49. Wave type that requires a medium? (Q49)

A) Light B) Radio C) Sound D) X-ray

Answer: C

50. Boiling point of water in Celsius? (Q50)

A) 0°C B) 50°C C) 100°C D) 212°C

Answer: C

51. SI unit of power? (Q51)

A) Joule B) Watt C) Volt D) Ampere

Answer: B

52. Type of lens thinner at edges, thicker in middle? (Q52)

A) Concave B) Convex C) Plano D) Cylindrical

Answer: B

53. Freezing point of water in Fahrenheit? (Q53)

A) 0°F B) 32°F C) 100°F D) 212°F

Answer: B

54. Color of light that bends most through a prism? (Q54)

A) Red B) Yellow C) Green D) Violet

Answer: D

55. SI unit of frequency? (Q55)

A) Second B) Hertz C) Meter D) Decibel

Answer: B

56. Which pole of a magnet attracts the north pole of another? (Q56)

A) North pole B) South pole C) Both D) Neither

Answer: B

57. What is the center of a lens called? (Q57)

A) Focus B) Optical center C) Pole D) Radius

Answer: B

58. Sound travels fastest in which medium? (Q58)

A) Air B) Water C) Steel D) Vacuum

Answer: C

59. What does 'g' stand for in physics equations? (Q59)

A) Gram B) Gravity acceleration C) Gauss D) Giga

Answer: B

60. Which mirror produces virtual and diminished images only? (Q60)

A) Plane B) Concave C) Convex D) Parabolic

Answer: C

61. What is the SI unit of force? (Q61)

A) Joule B) Newton C) Watt D) Pascal

Answer: B

62. Which is a scalar quantity? (Q62)

A) Velocity B) Force C) Speed D) Acceleration

Answer: C

63. Value of acceleration due to gravity (g)? (Q63)

A) 8.8 m/s² B) 9.8 m/s² C) 10.5 m/s² D) 11.2 m/s²

Answer: B

64. Instrument used to measure electric current? (Q64)

- A) Voltmeter B) Ammeter C) Ohmmeter D) Galvanometer

Answer: B

65. Energy possessed by a body due to motion? (Q65)

- A) Potential B) Kinetic C) Thermal D) Chemical

Answer: B

66. Newton's third law states action and reaction are: (Q66)

- A) Equal & same B) Unequal & opposite C) Equal & opposite D) Zero

Answer: C

67. SI unit of electrical resistance? (Q67)

- A) Volt B) Ampere C) Ohm D) Watt

Answer: C

68. Device converting electrical energy to mechanical? (Q68)

- A) Generator B) Motor C) Transformer D) Resistor

Answer: B

69. Wave type that requires a medium? (Q69)

- A) Light B) Radio C) Sound D) X-ray

Answer: C

70. Boiling point of water in Celsius? (Q70)

- A) 0°C B) 50°C C) 100°C D) 212°C

Answer: C

71. SI unit of power? (Q71)

- A) Joule B) Watt C) Volt D) Ampere

Answer: B

72. Type of lens thinner at edges, thicker in middle? (Q72)

- A) Concave B) Convex C) Plano D) Cylindrical

Answer: B

73. Freezing point of water in Fahrenheit? (Q73)

- A) 0°F B) 32°F C) 100°F D) 212°F

Answer: B

74. Color of light that bends most through a prism? (Q74)

- A) Red B) Yellow C) Green D) Violet

Answer: D

75. SI unit of frequency? (Q75)

- A) Second B) Hertz C) Meter D) Decibel

Answer: B

76. Which pole of a magnet attracts the north pole of another? (Q76)

- A) North pole B) South pole C) Both D) Neither

Answer: B

77. What is the center of a lens called? (Q77)

- A) Focus B) Optical center C) Pole D) Radius

Answer: B

78. Sound travels fastest in which medium? (Q78)

- A) Air B) Water C) Steel D) Vacuum

Answer: C

79. What does 'g' stand for in physics equations? (Q79)

- A) Gram B) Gravity acceleration C) Gauss D) Giga

Answer: B

80. Which mirror produces virtual and diminished images only? (Q80)

- A) Plane B) Concave C) Convex D) Parabolic

Answer: C

81. What is the SI unit of force? (Q81)

- A) Joule B) Newton C) Watt D) Pascal

Answer: B

82. Which is a scalar quantity? (Q82)

- A) Velocity B) Force C) Speed D) Acceleration

Answer: C

83. Value of acceleration due to gravity (g)? (Q83)

- A) 8.8 m/s^2 B) 9.8 m/s^2 C) 10.5 m/s^2 D) 11.2 m/s^2

Answer: B

84. Instrument used to measure electric current? (Q84)

- A) Voltmeter B) Ammeter C) Ohmmeter D) Galvanometer

Answer: B

85. Energy possessed by a body due to motion? (Q85)

- A) Potential B) Kinetic C) Thermal D) Chemical

Answer: B

86. Newton's third law states action and reaction are: (Q86)

- A) Equal & same B) Unequal & opposite C) Equal & opposite D) Zero

Answer: C

87. SI unit of electrical resistance? (Q87)

- A) Volt B) Ampere C) Ohm D) Watt

Answer: C

88. Device converting electrical energy to mechanical? (Q88)

- A) Generator B) Motor C) Transformer D) Resistor

Answer: B

89. Wave type that requires a medium? (Q89)

- A) Light B) Radio C) Sound D) X-ray

Answer: C

90. Boiling point of water in Celsius? (Q90)

- A) 0°C B) 50°C C) 100°C D) 212°C

Answer: C

91. SI unit of power? (Q91)

- A) Joule B) Watt C) Volt D) Ampere

Answer: B

92. Type of lens thinner at edges, thicker in middle? (Q92)

- A) Concave B) Convex C) Plano D) Cylindrical

Answer: B

93. Freezing point of water in Fahrenheit? (Q93)

- A) 0°F B) 32°F C) 100°F D) 212°F

Answer: B

94. Color of light that bends most through a prism? (Q94)

- A) Red B) Yellow C) Green D) Violet

Answer: D

95. SI unit of frequency? (Q95)

- A) Second B) Hertz C) Meter D) Decibel

Answer: B

96. Which pole of a magnet attracts the north pole of another? (Q96)

- A) North pole B) South pole C) Both D) Neither

Answer: B

97. What is the center of a lens called? (Q97)

- A) Focus B) Optical center C) Pole D) Radius

Answer: B

98. Sound travels fastest in which medium? (Q98)

- A) Air B) Water C) Steel D) Vacuum

Answer: C

99. What does 'g' stand for in physics equations? (Q99)

- A) Gram B) Gravity acceleration C) Gauss D) Giga

Answer: B

100. Which mirror produces virtual and diminished images only? (Q100)

- A) Plane B) Concave C) Convex D) Parabolic

Answer: C

Part 2: Normal Questions (101 - 200)

1. A car accelerates from rest at 2 m/s^2 for 5 s. What is its final velocity? (Q1)

- A) 5 m/s B) 10 m/s C) 15 m/s D) 20 m/s

Answer: B

2. What is the kinetic energy of a 2 kg mass moving at 4 m/s? (Q2)

- A) 8 J B) 16 J C) 32 J D) 64 J

Answer: B

3. Calculate the work done when a force of 10 N moves an object 5 m in the direction of force. (Q3)

- A) 2 J B) 25 J C) 50 J D) 100 J

Answer: C

4. What is the resistance of a circuit with 12 V voltage and 3 A current? (Q4)

- A) 2Ω B) 4Ω C) 6Ω D) 36Ω

Answer: B

5. Find the momentum of a 3 kg object traveling at 6 m/s. (Q5)

- A) $9 \text{ kg}\cdot\text{m/s}$ B) $18 \text{ kg}\cdot\text{m/s}$ C) $36 \text{ kg}\cdot\text{m/s}$ D) $54 \text{ kg}\cdot\text{m/s}$

Answer: B

6. What is the period of a wave with a frequency of 50 Hz? (Q6)

- A) 0.02 s B) 0.2 s C) 2 s D) 50 s

Answer: A

7. Calculate the pressure exerted by a force of 200 N on an area of 4 m^2 . (Q7)

- A) 25 Pa B) 50 Pa C) 100 Pa D) 800 Pa

Answer: B

8. What is the focal length of a convex mirror with a radius of curvature of 40 cm? (Q8)

- A) -20 cm B) +20 cm C) +40 cm D) -40 cm

Answer: B

9. How much heat is required to raise 1 kg of water by 10°C ? ($c = 4200 \text{ J/kg}\cdot\text{K}$) (Q9)

- A) 420 J B) 4200 J C) 42000 J D) 420000 J

Answer: C

10. An object is dropped from a height of 20 m. Find its time of fall ($g = 10 \text{ m/s}^2$). (Q10)

- A) 1 s B) 2 s C) 4 s D) 10 s

Answer: B

11. What power is consumed by a $100\ \Omega$ resistor carrying 2 A current? (Q11)

- A) 200 W B) 400 W C) W D) 20000 W

Answer: B

12. Calculate the buoyant force on a submerged body displacing $0.05\ \text{m}^3$ of water (density = $1000\ \text{kg/m}^3$). (Q12)

- A) 50 N B) 100 N C) 490 N D) 500 N

Answer: D

13. What is the wavelength of a sound wave in air having frequency 340 Hz and speed 340 m/s? (Q13)

- A) 0.5 m B) 1 m C) 2 m D) 340 m

Answer: B

14. Find the torque produced by a 50 N force applied perpendicularly 0.4 m from a pivot. (Q14)

- A) 10 N·m B) 20 N·m C) 125 N·m D) 200 N·m

Answer: B

15. What is the electric potential energy of two 2 C charges separated by 1 meter in vacuum? (Q15)

- A) $9 \times 10^9\ \text{J}$ B) $1.8 \times 10^{10}\ \text{J}$ C) $3.6 \times 10^{10}\ \text{J}$ D) Zero

Answer: C

16. Calculate the speed of light in a medium with refractive index 1.5 ($c = 3 \times 10^8\ \text{m/s}$). (Q16)

- A) $1.5 \times 10^8\ \text{m/s}$ B) $2 \times 10^8\ \text{m/s}$ C) $3 \times 10^8\ \text{m/s}$ D) $4.5 \times 10^8\ \text{m/s}$

Answer: B

17. What is the terminal velocity condition? (Q17)

- A) Acceleration is zero B) Velocity is zero C) Force of gravity is zero D) Drag is zero

Answer: A

18. Find the spring potential energy for a spring with constant $k = 200\ \text{N/m}$ stretched by 0.1 m. (Q18)

- A) 1 J B) 2 J C) 10 J D) 20 J

Answer: A

19. A transformer has 200 primary turns and 800 secondary turns. If input voltage is 120 V, output voltage is: (Q19)

- A) 30 V B) 240 V C) 480 V D) 960 V

Answer: C

20. What is the angular velocity of a wheel rotating at 120 revolutions per minute (rpm)? (Q20)

- A) $2\pi\ \text{rad/s}$ B) $4\pi\ \text{rad/s}$ C) $6\pi\ \text{rad/s}$ D) $120\pi\ \text{rad/s}$

Answer: B

21. A car accelerates from rest at $2\ \text{m/s}^2$ for 5 s. What is its final velocity? (Q21)

- A) 5 m/s B) 10 m/s C) 15 m/s D) 20 m/s

Answer: B

22. What is the kinetic energy of a 2 kg mass moving at 4 m/s? (Q22)

- A) 8 J B) 16 J C) 32 J D) 64 J

Answer: B

23. Calculate the work done when a force of 10 N moves an object 5 m in the direction of force. (Q23)

- A) 2 J B) 25 J C) 50 J D) 100 J

Answer: C

24. What is the resistance of a circuit with 12 V voltage and 3 A current? (Q24)

- A) $2\ \Omega$ B) $4\ \Omega$ C) $6\ \Omega$ D) $36\ \Omega$

Answer: B

25. Find the momentum of a 3 kg object traveling at 6 m/s. (Q25)

- A) $9\ \text{kg}\cdot\text{m/s}$ B) $18\ \text{kg}\cdot\text{m/s}$ C) $36\ \text{kg}\cdot\text{m/s}$ D) $54\ \text{kg}\cdot\text{m/s}$

Answer: B

26. What is the period of a wave with a frequency of 50 Hz? (Q26)

- A) 0.02 s B) 0.2 s C) 2 s D) 50 s

Answer: A

27. Calculate the pressure exerted by a force of 200 N on an area of 4 m². (Q27)

- A) 25 Pa B) 50 Pa C) 100 Pa D) 800 Pa

Answer: B

28. What is the focal length of a convex mirror with a radius of curvature of 40 cm? (Q28)

- A) -20 cm B) +20 cm C) +40 cm D) -40 cm

Answer: B

29. How much heat is required to raise 1 kg of water by 10 °C? ($c = 4200 \text{ J/kg}\cdot\text{K}$) (Q29)

- A) 420 J B) 4200 J C) 42000 J D) 420000 J

Answer: C

30. An object is dropped from a height of 20 m. Find its time of fall ($g = 10 \text{ m/s}^2$). (Q30)

- A) 1 s B) 2 s C) 4 s D) 10 s

Answer: B

31. What power is consumed by a 100 Ω resistor carrying 2 A current? (Q31)

- A) 200 W B) 400 W C) W D) 20000 W

Answer: B

32. Calculate the buoyant force on a submerged body displacing 0.05 m³ of water (density = 1000 kg/m³). (Q32)

- A) 50 N B) 100 N C) 490 N D) 500 N

Answer: D

33. What is the wavelength of a sound wave in air having frequency 340 Hz and speed 340 m/s? (Q33)

- A) 0.5 m B) 1 m C) 2 m D) 340 m

Answer: B

34. Find the torque produced by a 50 N force applied perpendicularly 0.4 m from a pivot. (Q34)

- A) 10 N·m B) 20 N·m C) 125 N·m D) 200 N·m

Answer: B

35. What is the electric potential energy of two 2 C charges separated by 1 meter in vacuum? (Q35)

- A) $9 \times 10^9 \text{ J}$ B) $1.8 \times 10^{10} \text{ J}$ C) $3.6 \times 10^{10} \text{ J}$ D) Zero

Answer: C

36. Calculate the speed of light in a medium with refractive index 1.5 ($c = 3 \times 10^8 \text{ m/s}$). (Q36)

- A) $1.5 \times 10^8 \text{ m/s}$ B) $2 \times 10^8 \text{ m/s}$ C) $3 \times 10^8 \text{ m/s}$ D) $4.5 \times 10^8 \text{ m/s}$

Answer: B

37. What is the terminal velocity condition? (Q37)

- A) Acceleration is zero B) Velocity is zero C) Force of gravity is zero D) Drag is zero

Answer: A

38. Find the spring potential energy for a spring with constant $k = 200 \text{ N/m}$ stretched by 0.1 m. (Q38)

- A) 1 J B) 2 J C) 10 J D) 20 J

Answer: A

39. A transformer has 200 primary turns and 800 secondary turns. If input voltage is 120 V, output voltage is: (Q39)

- A) 30 V B) 240 V C) 480 V D) 960 V

Answer: C

40. What is the angular velocity of a wheel rotating at 120 revolutions per minute (rpm)? (Q40)

- A) $2\pi \text{ rad/s}$ B) $4\pi \text{ rad/s}$ C) $6\pi \text{ rad/s}$ D) $120\pi \text{ rad/s}$

Answer: B

41. A car accelerates from rest at 2 m/s^2 for 5 s. What is its final velocity? (Q41)

- A) 5 m/s B) 10 m/s C) 15 m/s D) 20 m/s

Answer: B

42. What is the kinetic energy of a 2 kg mass moving at 4 m/s? (Q42)

- A) 8 J B) 16 J C) 32 J D) 64 J

Answer: B

43. Calculate the work done when a force of 10 N moves an object 5 m in the direction of force. (Q43)
A) 2 J B) 25 J C) 50 J D) 100 J
Answer: C
44. What is the resistance of a circuit with 12 V voltage and 3 A current? (Q44)
A) 2 Ω B) 4 Ω C) 6 Ω D) 36 Ω
Answer: B
45. Find the momentum of a 3 kg object traveling at 6 m/s. (Q45)
A) 9 kg·m/s B) 18 kg·m/s C) 36 kg·m/s D) 54 kg·m/s
Answer: B
46. What is the period of a wave with a frequency of 50 Hz? (Q46)
A) 0.02 s B) 0.2 s C) 2 s D) 50 s
Answer: A
47. Calculate the pressure exerted by a force of 200 N on an area of 4 m². (Q47)
A) 25 Pa B) 50 Pa C) 100 Pa D) 800 Pa
Answer: B
48. What is the focal length of a convex mirror with a radius of curvature of 40 cm? (Q48)
A) -20 cm B) +20 cm C) +40 cm D) -40 cm
Answer: B
49. How much heat is required to raise 1 kg of water by 10 °C? (c = 4200 J/kg·K) (Q49)
A) 420 J B) 4200 J C) 42000 J D) 420000 J
Answer: C
50. An object is dropped from a height of 20 m. Find its time of fall (g = 10 m/s²). (Q50)
A) 1 s B) 2 s C) 4 s D) 10 s
Answer: B
51. What power is consumed by a 100 Ω resistor carrying 2 A current? (Q51)
A) 200 W B) 400 W C) W D) 20000 W
Answer: B
52. Calculate the buoyant force on a submerged body displacing 0.05 m³ of water (density = 1000 kg/m³). (Q52)
A) 50 N B) 100 N C) 490 N D) 500 N
Answer: D
53. What is the wavelength of a sound wave in air having frequency 340 Hz and speed 340 m/s? (Q53)
A) 0.5 m B) 1 m C) 2 m D) 340 m
Answer: B
54. Find the torque produced by a 50 N force applied perpendicularly 0.4 m from a pivot. (Q54)
A) 10 N·m B) 20 N·m C) 125 N·m D) 200 N·m
Answer: B
55. What is the electric potential energy of two 2 C charges separated by 1 meter in vacuum? (Q55)
A) 9×10^9 J B) 1.8×10^{10} J C) 3.6×10^{10} J D) Zero
Answer: C
56. Calculate the speed of light in a medium with refractive index 1.5 (c = 3×10^8 m/s). (Q56)
A) 1.5×10^8 m/s B) 2×10^8 m/s C) 3×10^8 m/s D) 4.5×10^8 m/s
Answer: B
57. What is the terminal velocity condition? (Q57)
A) Acceleration is zero B) Velocity is zero C) Force of gravity is zero D) Drag is zero
Answer: A
58. Find the spring potential energy for a spring with constant k = 200 N/m stretched by 0.1 m. (Q58)
A) 1 J B) 2 J C) 10 J D) 20 J
Answer: A

59. A transformer has 200 primary turns and 800 secondary turns. If input voltage is 120 V, output voltage is: (Q59)
A) 30 V B) 240 V C) 480 V D) 960 V
Answer: C
60. What is the angular velocity of a wheel rotating at 120 revolutions per minute (rpm)? (Q60)
A) 2π rad/s B) 4π rad/s C) 6π rad/s D) 120π rad/s
Answer: B
61. A car accelerates from rest at 2 m/s^2 for 5 s. What is its final velocity? (Q61)
A) 5 m/s B) 10 m/s C) 15 m/s D) 20 m/s
Answer: B
62. What is the kinetic energy of a 2 kg mass moving at 4 m/s? (Q62)
A) 8 J B) 16 J C) 32 J D) 64 J
Answer: B
63. Calculate the work done when a force of 10 N moves an object 5 m in the direction of force. (Q63)
A) 2 J B) 25 J C) 50 J D) 100 J
Answer: C
64. What is the resistance of a circuit with 12 V voltage and 3 A current? (Q64)
A) 2Ω B) 4Ω C) 6Ω D) 36Ω
Answer: B
65. Find the momentum of a 3 kg object traveling at 6 m/s. (Q65)
A) $9 \text{ kg}\cdot\text{m/s}$ B) $18 \text{ kg}\cdot\text{m/s}$ C) $36 \text{ kg}\cdot\text{m/s}$ D) $54 \text{ kg}\cdot\text{m/s}$
Answer: B
66. What is the period of a wave with a frequency of 50 Hz? (Q66)
A) 0.02 s B) 0.2 s C) 2 s D) 50 s
Answer: A
67. Calculate the pressure exerted by a force of 200 N on an area of 4 m^2 . (Q67)
A) 25 Pa B) 50 Pa C) 100 Pa D) 800 Pa
Answer: B
68. What is the focal length of a convex mirror with a radius of curvature of 40 cm? (Q68)
A) -20 cm B) +20 cm C) +40 cm D) -40 cm
Answer: B
69. How much heat is required to raise 1 kg of water by 10°C ? ($c = 4200 \text{ J/kg}\cdot\text{K}$) (Q69)
A) 420 J B) 4200 J C) 42000 J D) 420000 J
Answer: C
70. An object is dropped from a height of 20 m. Find its time of fall ($g = 10 \text{ m/s}^2$). (Q70)
A) 1 s B) 2 s C) 4 s D) 10 s
Answer: B
71. What power is consumed by a 100Ω resistor carrying 2 A current? (Q71)
A) 200 W B) 400 W C) W D) 20000 W
Answer: B
72. Calculate the buoyant force on a submerged body displacing 0.05 m^3 of water (density = 1000 kg/m^3). (Q72)
A) 50 N B) 100 N C) 490 N D) 500 N
Answer: D
73. What is the wavelength of a sound wave in air having frequency 340 Hz and speed 340 m/s? (Q73)
A) 0.5 m B) 1 m C) 2 m D) 340 m
Answer: B
74. Find the torque produced by a 50 N force applied perpendicularly 0.4 m from a pivot. (Q74)
A) $10 \text{ N}\cdot\text{m}$ B) $20 \text{ N}\cdot\text{m}$ C) $125 \text{ N}\cdot\text{m}$ D) $200 \text{ N}\cdot\text{m}$
Answer: B

75. What is the electric potential energy of two 2 C charges separated by 1 meter in vacuum? (Q75)

- A) 9×10^9 J B) 1.8×10^{10} J C) 3.6×10^{10} J D) Zero

Answer: C

76. Calculate the speed of light in a medium with refractive index 1.5 ($c = 3 \times 10^8$ m/s). (Q76)

- A) 1.5×10^8 m/s B) 2×10^8 m/s C) 3×10^8 m/s D) 4.5×10^8 m/s

Answer: B

77. What is the terminal velocity condition? (Q77)

- A) Acceleration is zero B) Velocity is zero C) Force of gravity is zero D) Drag is zero

Answer: A

78. Find the spring potential energy for a spring with constant $k = 200$ N/m stretched by 0.1 m. (Q78)

- A) 1 J B) 2 J C) 10 J D) 20 J

Answer: A

79. A transformer has 200 primary turns and 800 secondary turns. If input voltage is 120 V, output voltage is: (Q79)

- A) 30 V B) 240 V C) 480 V D) 960 V

Answer: C

80. What is the angular velocity of a wheel rotating at 120 revolutions per minute (rpm)? (Q80)

- A) 2π rad/s B) 4π rad/s C) 6π rad/s D) 120π rad/s

Answer: B

81. A car accelerates from rest at 2 m/s^2 for 5 s. What is its final velocity? (Q81)

- A) 5 m/s B) 10 m/s C) 15 m/s D) 20 m/s

Answer: B

82. What is the kinetic energy of a 2 kg mass moving at 4 m/s? (Q82)

- A) 8 J B) 16 J C) 32 J D) 64 J

Answer: B

83. Calculate the work done when a force of 10 N moves an object 5 m in the direction of force. (Q83)

- A) 2 J B) 25 J C) 50 J D) 100 J

Answer: C

84. What is the resistance of a circuit with 12 V voltage and 3 A current? (Q84)

- A) 2Ω B) 4Ω C) 6Ω D) 36Ω

Answer: B

85. Find the momentum of a 3 kg object traveling at 6 m/s. (Q85)

- A) $9 \text{ kg}\cdot\text{m/s}$ B) $18 \text{ kg}\cdot\text{m/s}$ C) $36 \text{ kg}\cdot\text{m/s}$ D) $54 \text{ kg}\cdot\text{m/s}$

Answer: B

86. What is the period of a wave with a frequency of 50 Hz? (Q86)

- A) 0.02 s B) 0.2 s C) 2 s D) 50 s

Answer: A

87. Calculate the pressure exerted by a force of 200 N on an area of 4 m^2 . (Q87)

- A) 25 Pa B) 50 Pa C) 100 Pa D) 800 Pa

Answer: B

88. What is the focal length of a convex mirror with a radius of curvature of 40 cm? (Q88)

- A) -20 cm B) +20 cm C) +40 cm D) -40 cm

Answer: B

89. How much heat is required to raise 1 kg of water by 10°C ? ($c = 4200 \text{ J/kg}\cdot\text{K}$) (Q89)

- A) 420 J B) 4200 J C) 42000 J D) 420000 J

Answer: C

90. An object is dropped from a height of 20 m. Find its time of fall ($g = 10 \text{ m/s}^2$). (Q90)

- A) 1 s B) 2 s C) 4 s D) 10 s

Answer: B

91. What power is consumed by a $100\ \Omega$ resistor carrying 2 A current? (Q91)

- A) 200 W B) 400 W C) W D) 20000 W

Answer: B

92. Calculate the buoyant force on a submerged body displacing $0.05\ \text{m}^3$ of water (density = $1000\ \text{kg/m}^3$). (Q92)

- A) 50 N B) 100 N C) 490 N D) 500 N

Answer: D

93. What is the wavelength of a sound wave in air having frequency 340 Hz and speed 340 m/s? (Q93)

- A) 0.5 m B) 1 m C) 2 m D) 340 m

Answer: B

94. Find the torque produced by a 50 N force applied perpendicularly 0.4 m from a pivot. (Q94)

- A) 10 N·m B) 20 N·m C) 125 N·m D) 200 N·m

Answer: B

95. What is the electric potential energy of two 2 C charges separated by 1 meter in vacuum? (Q95)

- A) $9 \times 10^9\ \text{J}$ B) $1.8 \times 10^{10}\ \text{J}$ C) $3.6 \times 10^{10}\ \text{J}$ D) Zero

Answer: C

96. Calculate the speed of light in a medium with refractive index 1.5 ($c = 3 \times 10^8\ \text{m/s}$). (Q96)

- A) $1.5 \times 10^8\ \text{m/s}$ B) $2 \times 10^8\ \text{m/s}$ C) $3 \times 10^8\ \text{m/s}$ D) $4.5 \times 10^8\ \text{m/s}$

Answer: B

97. What is the terminal velocity condition? (Q97)

- A) Acceleration is zero B) Velocity is zero C) Force of gravity is zero D) Drag is zero

Answer: A

98. Find the spring potential energy for a spring with constant $k = 200\ \text{N/m}$ stretched by 0.1 m. (Q98)

- A) 1 J B) 2 J C) 10 J D) 20 J

Answer: A

99. A transformer has 200 primary turns and 800 secondary turns. If input voltage is 120 V, output voltage is: (Q99)

- A) 30 V B) 240 V C) 480 V D) 960 V

Answer: C

100. What is the angular velocity of a wheel rotating at 120 revolutions per minute (rpm)? (Q100)

- A) $2\pi\ \text{rad/s}$ B) $4\pi\ \text{rad/s}$ C) $6\pi\ \text{rad/s}$ D) $120\pi\ \text{rad/s}$

Answer: B

Part 3: Hard Questions (201 - 300)

1. An electron moves in a circular orbit of radius $0.5\ \text{\AA}$ in a magnetic field of 0.1 T perpendicular to velocity. What is its cyclotron frequency? (Q1)

- A) $1.76 \times 10^9\ \text{Hz}$ B) $2.8 \times 10^9\ \text{Hz}$ C) $1.6 \times 10^{10}\ \text{Hz}$ D) $2.8 \times 10^{10}\ \text{Hz}$

Answer: D

2. A particle of mass m is in a 1D box of length L . What is the ground state energy? (Q2)

- A) $h^2/(8mL^2)$ B) $h^2/(4mL^2)$ C) $h^2/(2mL^2)$ D) $h^2/(mL^2)$

Answer: A

3. What is the relativistic momentum of an electron moving at $0.8c$? (rest mass m_0) (Q3)

- A) $0.8\ m_0c$ B) $1.33\ m_0c$ C) $1.67\ m_0c$ D) $2.0\ m_0c$

Answer: B

4. A Carnot engine operates between 500 K and 300 K. What is its maximum efficiency? (Q4)

- A) 20% B) 40% C) 60% D) 80%

Answer: B

5. What is the de Broglie wavelength of a proton accelerated through a potential difference of 100 V? (Q5)

- A) 0.0286 Å B) 0.286 Å C) 2.86 Å D) 28.6 Å

Answer: A

6. An LRC series circuit has $R = 40 \, \Omega$, $L = 0.2 \, \text{H}$, and $C = 10 \, \mu\text{F}$ connected to a 100 V, 50 Hz source. What is the impedance Z? (Q6)

- A) 41.3 Ω B) 55.4 Ω C) 78.2 Ω D) 100 Ω

Answer: B

7. Calculate the Schwarzschild radius of a star with mass equal to 3 solar masses ($M_{\text{sun}} = 2 \times 10^{30} \, \text{kg}$). (Q7)

- A) 2.9 km B) 8.8 km C) 14.7 km D) 29.5 km

Answer: B

8. The wave function for a free particle is $\psi(x) = A \exp(ikx)$. What is the probability current density? (Q8)

- A) $\hbar k |A|^2 / m$ B) $\hbar k |A|^2 / (2m)$ C) 0 D) $k |A|^2 / m$

Answer: A

9. A uniform solid cylinder of mass M and radius R rolls without slipping down an inclined plane of angle θ . What is its linear acceleration? (Q9)

- A) $g \sin \theta$ B) $(2/3)g \sin \theta$ C) $(3/4)g \sin \theta$ D) $(1/2)g \sin \theta$

Answer: B

10. What is the magnetic field at the center of a circular loop of radius R carrying current I? (Q10)

- A) $\mu_0 I / (2R)$ B) $\mu_0 I / (4R)$ C) $2\mu_0 I / R$ D) $\mu_0 I / (\pi R)$

Answer: A

11. In a Young's double slit experiment, wavelength is 500 nm, slit separation is 1 mm, and screen distance is 2 m. What is the fringe width? (Q11)

- A) 0.5 mm B) 1.0 mm C) 1.5 mm D) 2.0 mm

Answer: B

12. What is the binding energy per nucleon for an alpha particle (mass of proton = 1.00727 u, neutron = 1.00866 u, alpha = 4.00150 u)? (Q12)

- A) 2.25 MeV B) 7.07 MeV C) 8.30 MeV D) 9.15 MeV

Answer: B

13. A parallel plate capacitor with air has capacitance C_0 . A dielectric slab of constant $K=4$ filling half the volume is inserted. New capacitance is: (Q13)

- A) 2.5 C_0 B) 3 C_0 C) 4 C_0 D) 5 C_0

Answer: A

14. The kinetic energy of a photon with wavelength 300 nm is approximately: (Q14)

- A) 2.07 eV B) 4.14 eV C) 6.63 eV D) 12.4 eV

Answer: B

15. Find the escape velocity from a planet of mass $6 \times 10^{24} \, \text{kg}$ and radius $6.4 \times 10^6 \, \text{m}$. (Q15)

- A) 7.9 km/s B) 11.2 km/s C) 21.4 km/s D) 42.1 km/s

Answer: B

16. A radioactive substance has a half-life of 4 hours. What fraction remains after 12 hours? (Q16)

- A) 1/2 B) 1/4 C) 1/8 D) 1/16

Answer: C

17. What is the Poynting vector magnitude for an electromagnetic wave with peak electric field $E = 300 \, \text{V/m}$? (Q17)

- A) 119 W/m^2 B) 238 W/m^2 C) 357 W/m^2 D) 476 W/m^2

Answer: B

18. An ideal gas undergoes isothermal expansion from volume V to 3V. What is the change in internal energy? (Q18)

- A) Zero B) $nRT \ln(3)$ C) $-nRT \ln(3)$ D) $3nRT$

Answer: A

19. What is the cutoff frequency for photoelectric emission from a metal with work function 2.5 eV? (Q19)

- A) 3.0×10^{14} Hz B) 6.0×10^{14} Hz C) 9.0×10^{14} Hz D) 1.2×10^{15} Hz

Answer: B

20. A rocket burns fuel at rate $dm/dt = 10$ kg/s with relative exhaust velocity $v_{ex} = 3000$ m/s. What is the thrust? (Q20)

- A) 1000 N B) 15000 N C) 30000 N D) 60000 N

Answer: C

21. An electron moves in a circular orbit of radius $0.5 \text{ \AA}$ in a magnetic field of 0.1 T perpendicular to velocity. What is its cyclotron frequency? (Q21)

- A) 1.76×10^9 Hz B) 2.8×10^9 Hz C) 1.6×10^{10} Hz D) 2.8×10^{10} Hz

Answer: D

22. A particle of mass m is in a 1D box of length L . What is the ground state energy? (Q22)

- A) $h^2/(8mL^2)$ B) $h^2/(4mL^2)$ C) $h^2/(2mL^2)$ D) $h^2/(mL^2)$

Answer: A

23. What is the relativistic momentum of an electron moving at $0.8c$? (rest mass m_0) (Q23)

- A) $0.8 m_0 c$ B) $1.33 m_0 c$ C) $1.67 m_0 c$ D) $2.0 m_0 c$

Answer: B

24. A Carnot engine operates between 500 K and 300 K. What is its maximum efficiency? (Q24)

- A) 20% B) 40% C) 60% D) 80%

Answer: B

25. What is the de Broglie wavelength of a proton accelerated through a potential difference of 100 V? (Q25)

- A) $0.0286 \text{ \AA}$ B) $0.286 \text{ \AA}$ C) $2.86 \text{ \AA}$ D) $28.6 \text{ \AA}$

Answer: A

26. An LRC series circuit has $R = 40 \Omega$, $L = 0.2$ H, and $C = 10 \mu\text{F}$ connected to a 100 V, 50 Hz source. What is the impedance Z ? (Q26)

- A) 41.3Ω B) 55.4Ω C) 78.2Ω D) 100Ω

Answer: B

27. Calculate the Schwarzschild radius of a star with mass equal to 3 solar masses ($M_{\text{sun}} = 2 \times 10^{30}$ kg). (Q27)

- A) 2.9 km B) 8.8 km C) 14.7 km D) 29.5 km

Answer: B

28. The wave function for a free particle is $\psi(x) = A \exp(ikx)$. What is the probability current density? (Q28)

- A) $\hbar k |A|^2 / m$ B) $\hbar k |A|^2 / (2m)$ C) 0 D) $k |A|^2 / m$

Answer: A

29. A uniform solid cylinder of mass M and radius R rolls without slipping down an inclined plane of angle θ . What is its linear acceleration? (Q29)

- A) $g \sin \theta$ B) $(2/3)g \sin \theta$ C) $(3/4)g \sin \theta$ D) $(1/2)g \sin \theta$

Answer: B

30. What is the magnetic field at the center of a circular loop of radius R carrying current I ? (Q30)

- A) $\mu_0 I / (2R)$ B) $\mu_0 I / (4R)$ C) $2\mu_0 I / R$ D) $\mu_0 I / (\pi R)$

Answer: A

31. In a Young's double slit experiment, wavelength is 500 nm, slit separation is 1 mm, and screen distance is 2 m. What is the fringe width? (Q31)

- A) 0.5 mm B) 1.0 mm C) 1.5 mm D) 2.0 mm

Answer: B

32. What is the binding energy per nucleon for an alpha particle (mass of proton = 1.00727 u, neutron = 1.00866 u, alpha = 4.00150 u)? (Q32)

- A) 2.25 MeV B) 7.07 MeV C) 8.30 MeV D) 9.15 MeV

Answer: B

33. A parallel plate capacitor with air has capacitance C_0 . A dielectric slab of constant $K=4$ filling half the volume is inserted. New capacitance is: (Q33)

- A) $2.5 C_0$ B) $3 C_0$ C) $4 C_0$ D) $5 C_0$

Answer: A

34. The kinetic energy of a photon with wavelength 300 nm is approximately: (Q34)

- A) 2.07 eV B) 4.14 eV C) 6.63 eV D) 12.4 eV

Answer: B

35. Find the escape velocity from a planet of mass 6×10^{24} kg and radius 6.4×10^6 m. (Q35)

- A) 7.9 km/s B) 11.2 km/s C) 21.4 km/s D) 42.1 km/s

Answer: B

36. A radioactive substance has a half-life of 4 hours. What fraction remains after 12 hours? (Q36)

- A) $1/2$ B) $1/4$ C) $1/8$ D) $1/16$

Answer: C

37. What is the Poynting vector magnitude for an electromagnetic wave with peak electric field $E = 300$ V/m? (Q37)

- A) 119 W/m^2 B) 238 W/m^2 C) 357 W/m^2 D) 476 W/m^2

Answer: B

38. An ideal gas undergoes isothermal expansion from volume V to $3V$. What is the change in internal energy? (Q38)

- A) Zero B) $nRT \ln(3)$ C) $-nRT \ln(3)$ D) $3nRT$

Answer: A

39. What is the cutoff frequency for photoelectric emission from a metal with work function 2.5 eV? (Q39)

- A) 3.0×10^{14} Hz B) 6.0×10^{14} Hz C) 9.0×10^{14} Hz D) 1.2×10^{15} Hz

Answer: B

40. A rocket burns fuel at rate $dm/dt = 10$ kg/s with relative exhaust velocity $v_{\text{ex}} = 3000$ m/s. What is the thrust? (Q40)

- A) 1000 N B) 15000 N C) 30000 N D) 60000 N

Answer: C

41. An electron moves in a circular orbit of radius $0.5 \text{ \AA}$ in a magnetic field of 0.1 T perpendicular to velocity. What is its cyclotron frequency? (Q41)

- A) 1.76×10^9 Hz B) 2.8×10^9 Hz C) 1.6×10^{10} Hz D) 2.8×10^{10} Hz

Answer: D

42. A particle of mass m is in a 1D box of length L . What is the ground state energy? (Q42)

- A) $h^2/(8mL^2)$ B) $h^2/(4mL^2)$ C) $h^2/(2mL^2)$ D) $h^2/(mL^2)$

Answer: A

43. What is the relativistic momentum of an electron moving at $0.8c$? (rest mass m_0) (Q43)

- A) $0.8 m_0 c$ B) $1.33 m_0 c$ C) $1.67 m_0 c$ D) $2.0 m_0 c$

Answer: B

44. A Carnot engine operates between 500 K and 300 K . What is its maximum efficiency? (Q44)

- A) 20% B) 40% C) 60% D) 80%

Answer: B

45. What is the de Broglie wavelength of a proton accelerated through a potential difference of 100 V ? (Q45)

- A) $0.0286 \text{ \AA}$ B) $0.286 \text{ \AA}$ C) $2.86 \text{ \AA}$ D) $28.6 \text{ \AA}$

Answer: A

46. An LRC series circuit has $R = 40 \text{ } \Omega$, $L = 0.2 \text{ H}$, and $C = 10 \text{ } \mu\text{F}$ connected to a 100 V , 50 Hz source. What is the impedance Z ? (Q46)

- A) $41.3 \text{ } \Omega$ B) $55.4 \text{ } \Omega$ C) $78.2 \text{ } \Omega$ D) $100 \text{ } \Omega$

Answer: B

47. Calculate the Schwarzschild radius of a star with mass equal to 3 solar masses ($M_{\text{sun}} = 2 \times 10^{30} \text{ kg}$). (Q47)

- A) 2.9 km B) 8.8 km C) 14.7 km D) 29.5 km

Answer: B

48. The wave function for a free particle is $\psi(x) = A \exp(ikx)$. What is the probability current density? (Q48)

- A) $\hbar k |A|^2 / m$ B) $\hbar k |A|^2 / (2m)$ C) 0 D) $k |A|^2 / m$

Answer: A

49. A uniform solid cylinder of mass M and radius R rolls without slipping down an inclined plane of angle θ .

What is its linear acceleration? (Q49)

- A) $g \sin \theta$ B) $(2/3)g \sin \theta$ C) $(3/4)g \sin \theta$ D) $(1/2)g \sin \theta$

Answer: B

50. What is the magnetic field at the center of a circular loop of radius R carrying current I ? (Q50)

- A) $\mu_0 I / (2R)$ B) $\mu_0 I / (4R)$ C) $2\mu_0 I / R$ D) $\mu_0 I / (\pi R)$

Answer: A

51. In a Young's double slit experiment, wavelength is 500 nm, slit separation is 1 mm, and screen distance is 2 m. What is the fringe width? (Q51)

- A) 0.5 mm B) 1.0 mm C) 1.5 mm D) 2.0 mm

Answer: B

52. What is the binding energy per nucleon for an alpha particle (mass of proton = 1.00727 u, neutron = 1.00866 u, alpha = 4.00150 u)? (Q52)

- A) 2.25 MeV B) 7.07 MeV C) 8.30 MeV D) 9.15 MeV

Answer: B

53. A parallel plate capacitor with air has capacitance C_0 . A dielectric slab of constant $K=4$ filling half the volume is inserted. New capacitance is: (Q53)

- A) $2.5 C_0$ B) $3 C_0$ C) $4 C_0$ D) $5 C_0$

Answer: A

54. The kinetic energy of a photon with wavelength 300 nm is approximately: (Q54)

- A) 2.07 eV B) 4.14 eV C) 6.63 eV D) 12.4 eV

Answer: B

55. Find the escape velocity from a planet of mass $6 \times 10^{24} \text{ kg}$ and radius $6.4 \times 10^6 \text{ m}$. (Q55)

- A) 7.9 km/s B) 11.2 km/s C) 21.4 km/s D) 42.1 km/s

Answer: B

56. A radioactive substance has a half-life of 4 hours. What fraction remains after 12 hours? (Q56)

- A) $1/2$ B) $1/4$ C) $1/8$ D) $1/16$

Answer: C

57. What is the Poynting vector magnitude for an electromagnetic wave with peak electric field $E = 300 \text{ V/m}$? (Q57)

- A) 119 W/m^2 B) 238 W/m^2 C) 357 W/m^2 D) 476 W/m^2

Answer: B

58. An ideal gas undergoes isothermal expansion from volume V to $3V$. What is the change in internal energy? (Q58)

- A) Zero B) $nRT \ln(3)$ C) $-nRT \ln(3)$ D) $3nRT$

Answer: A

59. What is the cutoff frequency for photoelectric emission from a metal with work function 2.5 eV? (Q59)

- A) $3.0 \times 10^{14} \text{ Hz}$ B) $6.0 \times 10^{14} \text{ Hz}$ C) $9.0 \times 10^{14} \text{ Hz}$ D) $1.2 \times 10^{15} \text{ Hz}$

Answer: B

60. A rocket burns fuel at rate $dm/dt = 10 \text{ kg/s}$ with relative exhaust velocity $v_{\text{ex}} = 3000 \text{ m/s}$. What is the thrust? (Q60)

- A) 1000 N B) 15000 N C) 30000 N D) 60000 N

Answer: C

61. An electron moves in a circular orbit of radius $0.5 \text{ \AA}$ in a magnetic field of 0.1 T perpendicular to velocity. What is its cyclotron frequency? (Q61)
 A) $1.76 \times 10^9 \text{ Hz}$ B) $2.8 \times 10^9 \text{ Hz}$ C) $1.6 \times 10^{10} \text{ Hz}$ D) $2.8 \times 10^{10} \text{ Hz}$
Answer: D
62. A particle of mass m is in a 1D box of length L . What is the ground state energy? (Q62)
 A) $h^2/(8mL^2)$ B) $h^2/(4mL^2)$ C) $h^2/(2mL^2)$ D) $h^2/(mL^2)$
Answer: A
63. What is the relativistic momentum of an electron moving at $0.8c$? (rest mass m_0) (Q63)
 A) $0.8 m_0 c$ B) $1.33 m_0 c$ C) $1.67 m_0 c$ D) $2.0 m_0 c$
Answer: B
64. A Carnot engine operates between 500 K and 300 K . What is its maximum efficiency? (Q64)
 A) 20% B) 40% C) 60% D) 80%
Answer: B
65. What is the de Broglie wavelength of a proton accelerated through a potential difference of 100 V ? (Q65)
 A) $0.0286 \text{ \AA}$ B) $0.286 \text{ \AA}$ C) $2.86 \text{ \AA}$ D) $28.6 \text{ \AA}$
Answer: A
66. An LRC series circuit has $R = 40 \text{ } \Omega$, $L = 0.2 \text{ H}$, and $C = 10 \text{ } \mu\text{F}$ connected to a 100 V , 50 Hz source. What is the impedance Z ? (Q66)
 A) $41.3 \text{ } \Omega$ B) $55.4 \text{ } \Omega$ C) $78.2 \text{ } \Omega$ D) $100 \text{ } \Omega$
Answer: B
67. Calculate the Schwarzschild radius of a star with mass equal to 3 solar masses ($M_{\text{sun}} = 2 \times 10^{30} \text{ kg}$). (Q67)
 A) 2.9 km B) 8.8 km C) 14.7 km D) 29.5 km
Answer: B
68. The wave function for a free particle is $\psi(x) = A \exp(ikx)$. What is the probability current density? (Q68)
 A) $\hbar k |A|^2 / m$ B) $\hbar k |A|^2 / (2m)$ C) 0 D) $k |A|^2 / m$
Answer: A
69. A uniform solid cylinder of mass M and radius R rolls without slipping down an inclined plane of angle θ . What is its linear acceleration? (Q69)
 A) $g \sin \theta$ B) $(2/3)g \sin \theta$ C) $(3/4)g \sin \theta$ D) $(1/2)g \sin \theta$
Answer: B
70. What is the magnetic field at the center of a circular loop of radius R carrying current I ? (Q70)
 A) $\mu_0 I / (2R)$ B) $\mu_0 I / (4R)$ C) $2\mu_0 I / R$ D) $\mu_0 I / (\pi R)$
Answer: A
71. In a Young's double slit experiment, wavelength is 500 nm , slit separation is 1 mm , and screen distance is 2 m . What is the fringe width? (Q71)
 A) 0.5 mm B) 1.0 mm C) 1.5 mm D) 2.0 mm
Answer: B
72. What is the binding energy per nucleon for an alpha particle (mass of proton = 1.00727 u , neutron = 1.00866 u , alpha = 4.00150 u)? (Q72)
 A) 2.25 MeV B) 7.07 MeV C) 8.30 MeV D) 9.15 MeV
Answer: B
73. A parallel plate capacitor with air has capacitance C_0 . A dielectric slab of constant $K=4$ filling half the volume is inserted. New capacitance is: (Q73)
 A) $2.5 C_0$ B) $3 C_0$ C) $4 C_0$ D) $5 C_0$
Answer: A
74. The kinetic energy of a photon with wavelength 300 nm is approximately: (Q74)
 A) 2.07 eV B) 4.14 eV C) 6.63 eV D) 12.4 eV
Answer: B

75. Find the escape velocity from a planet of mass 6×10^{24} kg and radius 6.4×10^6 m. (Q75)

- A) 7.9 km/s B) 11.2 km/s C) 21.4 km/s D) 42.1 km/s

Answer: B

76. A radioactive substance has a half-life of 4 hours. What fraction remains after 12 hours? (Q76)

- A) $1/2$ B) $1/4$ C) $1/8$ D) $1/16$

Answer: C

77. What is the Poynting vector magnitude for an electromagnetic wave with peak electric field $E = 300$ V/m? (Q77)

- A) 119 W/m² B) 238 W/m² C) 357 W/m² D) 476 W/m²

Answer: B

78. An ideal gas undergoes isothermal expansion from volume V to $3V$. What is the change in internal energy? (Q78)

- A) Zero B) $nRT \ln(3)$ C) $-nRT \ln(3)$ D) $3nRT$

Answer: A

79. What is the cutoff frequency for photoelectric emission from a metal with work function 2.5 eV? (Q79)

- A) 3.0×10^{14} Hz B) 6.0×10^{14} Hz C) 9.0×10^{14} Hz D) 1.2×10^{15} Hz

Answer: B

80. A rocket burns fuel at rate $dm/dt = 10$ kg/s with relative exhaust velocity $v_{ex} = 3000$ m/s. What is the thrust? (Q80)

- A) 1000 N B) 15000 N C) 30000 N D) 60000 N

Answer: C

81. An electron moves in a circular orbit of radius $0.5 \text{ \AA}$ in a magnetic field of 0.1 T perpendicular to velocity. What is its cyclotron frequency? (Q81)

- A) 1.76×10^9 Hz B) 2.8×10^9 Hz C) 1.6×10^{10} Hz D) 2.8×10^{10} Hz

Answer: D

82. A particle of mass m is in a 1D box of length L . What is the ground state energy? (Q82)

- A) $h^2/(8mL^2)$ B) $h^2/(4mL^2)$ C) $h^2/(2mL^2)$ D) $h^2/(mL^2)$

Answer: A

83. What is the relativistic momentum of an electron moving at $0.8c$? (rest mass m_0) (Q83)

- A) $0.8 m_0 c$ B) $1.33 m_0 c$ C) $1.67 m_0 c$ D) $2.0 m_0 c$

Answer: B

84. A Carnot engine operates between 500 K and 300 K. What is its maximum efficiency? (Q84)

- A) 20% B) 40% C) 60% D) 80%

Answer: B

85. What is the de Broglie wavelength of a proton accelerated through a potential difference of 100 V? (Q85)

- A) $0.0286 \text{ \AA}$ B) $0.286 \text{ \AA}$ C) $2.86 \text{ \AA}$ D) $28.6 \text{ \AA}$

Answer: A

86. An LRC series circuit has $R = 40 \text{ } \Omega$, $L = 0.2$ H, and $C = 10 \text{ } \mu\text{F}$ connected to a 100 V, 50 Hz source. What is the impedance Z ? (Q86)

- A) $41.3 \text{ } \Omega$ B) $55.4 \text{ } \Omega$ C) $78.2 \text{ } \Omega$ D) $100 \text{ } \Omega$

Answer: B

87. Calculate the Schwarzschild radius of a star with mass equal to 3 solar masses ($M_{\text{sun}} = 2 \times 10^{30}$ kg). (Q87)

- A) 2.9 km B) 8.8 km C) 14.7 km D) 29.5 km

Answer: B

88. The wave function for a free particle is $\psi(x) = A \exp(ikx)$. What is the probability current density? (Q88)

- A) $\hbar k |A|^2 / m$ B) $\hbar k |A|^2 / (2m)$ C) 0 D) $k |A|^2 / m$

Answer: A

89. A uniform solid cylinder of mass M and radius R rolls without slipping down an inclined plane of angle θ . What is its linear acceleration? (Q89)
 A) $g \sin \theta$ B) $(2/3)g \sin \theta$ C) $(3/4)g \sin \theta$ D) $(1/2)g \sin \theta$
Answer: B
90. What is the magnetic field at the center of a circular loop of radius R carrying current I ? (Q90)
 A) $\mu_0 I / (2R)$ B) $\mu_0 I / (4R)$ C) $2\mu_0 I / R$ D) $\mu_0 I / (\pi R)$
Answer: A
91. In a Young's double slit experiment, wavelength is 500 nm, slit separation is 1 mm, and screen distance is 2 m. What is the fringe width? (Q91)
 A) 0.5 mm B) 1.0 mm C) 1.5 mm D) 2.0 mm
Answer: B
92. What is the binding energy per nucleon for an alpha particle (mass of proton = 1.00727 u, neutron = 1.00866 u, alpha = 4.00150 u)? (Q92)
 A) 2.25 MeV B) 7.07 MeV C) 8.30 MeV D) 9.15 MeV
Answer: B
93. A parallel plate capacitor with air has capacitance C_0 . A dielectric slab of constant $K=4$ filling half the volume is inserted. New capacitance is: (Q93)
 A) $2.5 C_0$ B) $3 C_0$ C) $4 C_0$ D) $5 C_0$
Answer: A
94. The kinetic energy of a photon with wavelength 300 nm is approximately: (Q94)
 A) 2.07 eV B) 4.14 eV C) 6.63 eV D) 12.4 eV
Answer: B
95. Find the escape velocity from a planet of mass 6×10^{24} kg and radius 6.4×10^6 m. (Q95)
 A) 7.9 km/s B) 11.2 km/s C) 21.4 km/s D) 42.1 km/s
Answer: B
96. A radioactive substance has a half-life of 4 hours. What fraction remains after 12 hours? (Q96)
 A) $1/2$ B) $1/4$ C) $1/8$ D) $1/16$
Answer: C
97. What is the Poynting vector magnitude for an electromagnetic wave with peak electric field $E = 300$ V/m? (Q97)
 A) 119 W/m² B) 238 W/m² C) 357 W/m² D) 476 W/m²
Answer: B
98. An ideal gas undergoes isothermal expansion from volume V to $3V$. What is the change in internal energy? (Q98)
 A) Zero B) $nRT \ln(3)$ C) $-nRT \ln(3)$ D) $3nRT$
Answer: A
99. What is the cutoff frequency for photoelectric emission from a metal with work function 2.5 eV? (Q99)
 A) 3.0×10^{14} Hz B) 6.0×10^{14} Hz C) 9.0×10^{14} Hz D) 1.2×10^{15} Hz
Answer: B
100. A rocket burns fuel at rate $dm/dt = 10$ kg/s with relative exhaust velocity $v_{ex} = 3000$ m/s. What is the thrust? (Q100)
 A) 1000 N B) 15000 N C) 30000 N D) 60000 N
Answer: C